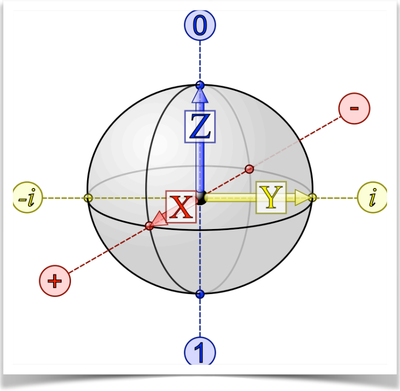

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IQC 2023/24 (756AA, 6CFU)

Introduction to Quantum Computing

Shor's algorithm for integer factorization

Shor's algorithm for integer factorization

- N composite number, of $n = \lceil \log_2 \mathrm{N} \rceil$ bits

  Shor's algorithm can find a factor of N in $O(n^2 \log n \log \log n)$ steps
  ➔ time polynomial in the input dimension n

- The most efficient classical factoring algorithm (GENERAL NUMBER FIELD SIEVE) works in subexponential time $O(e^{1.9\, n^{1/3} \log^{2/3} n})$

- The decision version of the problem is **not** NP-complete
  - it could be in P, with P ≠ NP (i.e., without implying the collapse of the polynomial gerarchy)

- Public key cryptography (RSA, DH, ECC) would be broken if Shor's algorithm could be physically realized (for big integers)

Complexity Classes

BPP: Bounded Error Probabilistic Polynomial time

Class of decision problems that can be solved in polynomial time using randomized algorithms, if a bounded probability of error is allowed in the solution

BPP is regarded as being, even more than P, the **class of problems that can be efficiently solved on a classical computer**

The (decision version of) Integer Factorization is **not known** to be in BPP

It is generally believed that **P = BPP**

The relationship between **BPP** and **NP** is unknown: it is not known whether **BPP** is a subset of **NP**, **NP** is a subset of **BPP** or neither

$$\boxed{\begin{array}{l} \mathbf{P} \subseteq \mathbf{BPP} \\ \mathbf{BPP}\ ?\ \mathbf{NP} \end{array}}$$

Complexity Classes

BQP: Bounded Error **Quantum** Polynomial time

Class of decision problems that can be solved in polynomial time on a **quantum computer**, with bounded error probability

The (decision version of) **Integer Factorization** belongs to **BQP**

If we could prove that (the decision version of)

Integer Factorization $\notin$ BPP

$\Rightarrow$

Quantum computers would provide the first counter-example to the "Strong" Church-Turing Thesis, which states that all reasonable models of computation are polynomially equivalent

$P \subseteq BPP \subseteq BQP \subseteq PSPACE$

It is believed that **P = BPP**, while the other inclusions are believed to be strict

$BPP \subseteq BQP$

a polynomial time randomized algorithms can be written as a polynomial-size reversible circuit that involves some coin flips

quantum computers can generate coin flips using Hadamard gates, then run the reversible circuit, and measure the final answer bit

A proof that BPP ≠ BQP (for instance, a proof that Integer Factorization ∉ BPP) would imply P ≠ PSPACE, solving one of the main open problems in computers science since the 1960s

$BQP \subseteq PSPACE$

A quantum system can be simulated efficiently in terms of space (though not necessarily in terms of time)

BQP vs NP

It is believed that NP-complete problems are probably not in BQP

An *evidence* is the lower bound for Grover search: a quantum brute-force search on all 2^n possible assignments to an n-variable formula gives a square-root speed-up, but not more

(of course, this not a proof)

There could also be problems in BQP that are not in NP

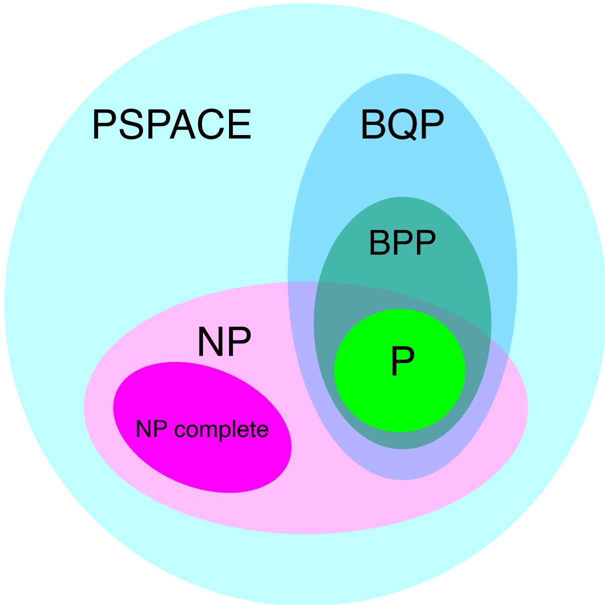
BQP and NP might be incomparable

P ⊆ NP ⊆ PSPACE

P ⊆ BPP ⊆ BQP ⊆ PSPACE

BPP ? NP

BQP ? NP



Shor's algorithm for integer factorization

INTEGER FACTORIZATION is solved via a reduction to ORDER FINDING

Integer Factorization reduces probabilistically to the Order Finding problem (Miller, 1975)

If we have an efficient algorithm for Order Finding, we can give an efficient algorithm for Factorization

ORDER FINDING is solved with a quantum algorithm, consisting in an application of quantum phase estimation (QPE)

Shor's algorithm thus consists of two parts:

- one **can be run on a classical computer** (reduction to ORDER FINDING))
- one **must be run on a quantum computer** (ORDER FINDING)

The Order Finding Problem

Definition (Order)

Given integers a and N, s.t. $GCD(a, N) = 1$, the ORDER of a is the **smallest positive integer** r such that

$$a^r \equiv 1 \mod N$$

Note

- The fact that a has a well-defined order r follows from **Euler's Theorem**, which states that

  $$a^{\phi(N)} \equiv 1 \mod N$$

  for any integers a and N, s.t. $GCD(a, N) = 1$

- $\phi(N)$ is the **Euler's totient function,** defined as the number of positive integers $\leq N$ and relatively prime to N

- Moreover, we can always assume that $r < N$ since

  $$r \leq \phi(N) \leq N - 1$$

The Order Finding Problem

Definition (Order)

Given integers a and N, s.t. $GCD(a, N) = 1$, the ORDER of a is the **smallest positive integer** r such that

$$a^r \equiv 1 \mod N$$

Note

The order r is also called PERIOD, since the function

$$f(x) = a^x \bmod N$$

is periodic, with period $r < N$

$$\forall x, \quad f(x + r) = f(x)$$

Indeed

$$\forall x, \quad f(x + r) = a^{x+r} \bmod N = a^x a^r \bmod N = a^x \bmod N = f(x)$$

$f(r) = a^r \bmod N = 1$

Examples

$N = 15$
$a = 2$

$r = 4$

x	0	1	2	3	**4**	5	6	7	8	⋯
$a^x \mod N$	1	2	4	8	**1**	2	4	8	1	⋯

After it hits 1, the sequence of numbers will simply repeat

$N = 15$
$a = 4$

$r = 2$

x	0	1	**2**	3	4	5	6	7	8	⋯
$a^x \mod N$	1	4	**1**	4	1	4	1	4	1	⋯

The Order Finding Problem

INPUT integer a and N, s.t. $GCD(a, N) = 1$

PROBLEM find the ORDER of a modulo N

Complexity

- Belivied to be hard on a classical computer
- No algorithm is known to solve the problem in time polynomial in the $n = \lceil \log N \rceil$ bits needed to specify the input
- Best known probabilistic classical algorithm has complexity $O\left(e^{\sqrt{n \log n}}\right)$

Quantum Phase Estimation can be used to find the order efficiently on a quantum computer

Quantum Phase Estimation (QPE)

Given a n-qubit unitary operator U, and an eigenvector $|\psi\rangle$ of U with corresponding **unknown** eigenvalue λ

$$U|\psi\rangle = \lambda|\psi\rangle$$

Compute, or at least approximate, the eigenvalue λ

Property

All eigenvalues of a unitary matrix U have modulus 1

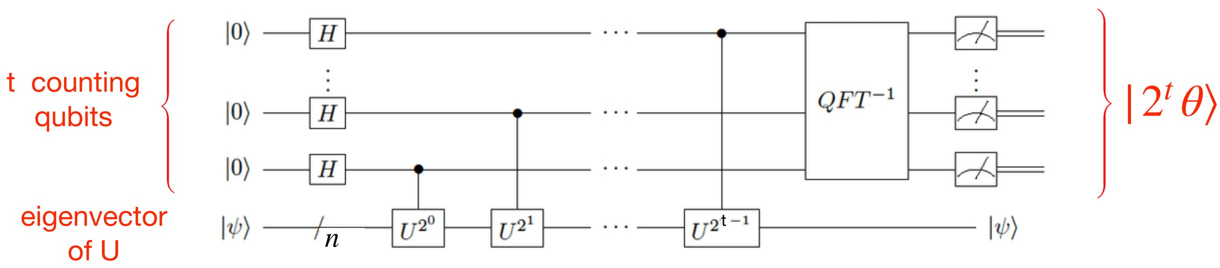
We can write the eigenvalue λ as $\lambda = e^{2\pi i\theta}$ for some real number $\theta \in [0,1)$

Thus, the only information that matters is the phase θ

Quantum Phase Estimation (QPE)

$$U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle$$

$|\psi\rangle$ is an eigenvector of U, $e^{2\pi i\theta}$ is the corresponding eigenvalue



The accuracy and the probability of success of QPE depend on the number t of counting qubits

We need to define a unitary operator U_a that computes

$$f(x) = a^x \bmod N$$

and whose eigenvalues contain the order *r*, so that we can apply QPE to estimate the eigenvalues, and recover the order

Construction of the operator U_a

$a \in \{2, \ldots, N-1\}$ $\quad GCD(a, N) = 1$ $\quad n = \lceil \log_2 N \rceil$

$$U_a |y\rangle = \begin{cases} |a\, y \bmod N\,\rangle & 0 \le y < N \\ |\,y\,\rangle & N \le y < 2^n \end{cases}$$

U_a acts non trivially when $0 \le y < N$, and is UNITARY

Example

$N = 5$ $\quad a = 3$

$n = \lceil \log_2 N \rceil = \lceil \log_2 5 \rceil = 3$

$$U_a |y\rangle = \begin{cases} |\,3\, y \bmod 5\,\rangle & 0 \le y < 5 \\ |\,y\,\rangle & 5 \le y < 8 \end{cases}$$

U_a is UNITARY

U_a is a permutation matrix

$U_a^\dagger U_a = I$

$$U_a = \begin{array}{c} \begin{pmatrix} U & 0 \\ 0 & I \end{pmatrix} \begin{array}{l} \} \; N \\ \} \; 2^n - N \end{array} \\ \quad \underbrace{}_{N} \quad \underbrace{}_{2^n - N} \end{array}$$

PROOF

It is enough to show that $U^\dagger U = I$

Let $i, j \in [0,\, N-1]$, and consider a generic entry $\langle i\,|\,U^\dagger U\,|\,j\rangle$ of $U^\dagger U$:

$$\langle i\,|\,U^\dagger U\,|\,j\rangle = \langle a\, i \mod N\,|\,a\, j \mod N\rangle$$

Since $|\,a\, i \mod N\rangle$ and $|\,a\, j \mod N\rangle$ are computational basis states, we have

$$\langle i\,|\,U^\dagger U\,|\,j\rangle = \langle a\, i \bmod N\,|\,a\, j \bmod N\rangle = 1 \text{ iff } ai \equiv aj \mod N$$

Thus, we have to prove that $ai \equiv aj \mod N$ iff $i = j$

Suppose that $ai \equiv aj \mod N$. Since a has an inverse ($GCD(a, N) = 1$), we get

$a^{-1}ai \equiv a^{-1}aj \mod N$

$i \equiv j \mod N$

$i = j$ since $i, j < N$ ∎

Eigenvalues of U_a

The eigenvalues of U_a are the r-roots of unity

$$U_a^r |y\rangle = U_a^{(r-1)} |a\,y \bmod N\rangle = U_a^{(r-2)} |a^2 y \bmod N\rangle = \dots$$
$$= U_a |a^{(r-1)} y \bmod N\rangle = |a^r y \bmod N\rangle = |y\rangle \qquad 0 \le y < N$$

$$U_a^r |y\rangle = |y\rangle \qquad N \le y < 2^n$$

$$\Rightarrow \boxed{U_a^r = I}$$

$$\left.\begin{array}{l} U_a |u\rangle = \lambda |u\rangle \Rightarrow U_a^r |u\rangle = \lambda^r |u\rangle \\ U_a^r |u\rangle = |u\rangle \end{array}\right\} \Rightarrow \boxed{\lambda^r = 1}$$

$$\lambda_s = e^{\frac{2\pi i}{r} s} = \omega^s \qquad \omega = e^{\frac{2\pi i}{r}} \qquad 0 \le s < r$$

Thus, the eigenvalues of U_a contain the order r of a

➔ we can use QPE to estimate the eigenvalues and recover the order r

➔ *Shor's solution is QPE on U_a*

Eigenvectors of U_a

$$U_a |y\rangle = \begin{cases} |a\,y \bmod N\rangle & 0 \le y < N \\ |y\rangle & N \le y < 2^n \end{cases}$$

For $0 \le s < r$, let $\omega = e^{\frac{2\pi i}{r}}$

Consider the state

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \omega^{-sk} |a^k \bmod N\rangle$$

$$U_a |u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \omega^{-sk} |a^{k+1} \bmod N\rangle = \frac{1}{\sqrt{r}} \sum_{k=1}^{r} \omega^{-s(k-1)} |a^k \bmod N\rangle$$

$$= \frac{1}{\sqrt{r}} \sum_{k=1}^{r-1} \omega^{-sk} \omega^s |a^k \bmod N\rangle + \frac{1}{\sqrt{r}} \omega^{-sr} \omega^s |a^r \bmod N\rangle$$

The term with $k = r$ is **equal** to the term with $k = 0$

- $a^r \bmod N = 1 = a^0 \bmod N$
- $\omega^{-sr} = (e^{\frac{2\pi i}{r}})^{-sr} = e^{-2\pi i s} = 1 = \omega^{-s0}$

Eigenvectors of U_a

$$U_a |y\rangle = \begin{cases} |a\,y \bmod N\rangle & 0 \le y < N \\ |y\rangle & N \le y < 2^n \end{cases}$$

For $0 \le s < r$, let $\omega = e^{\frac{2\pi i}{r}}$

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$$= \frac{1}{\sqrt{r}} \sum_{k=1}^{r-1} \omega^{-sk} \omega^s |a^k \bmod N\rangle + \frac{1}{\sqrt{r}} \omega^{-sr} \omega^s |a^r \bmod N\rangle$$

$$= \frac{1}{\sqrt{r}} \sum_{k=1}^{r-1} \omega^{-sk} \omega^s |a^k \bmod N\rangle + \frac{1}{\sqrt{r}} \omega^{-s0} \omega^s |a^0 \bmod N\rangle$$

$$= \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \omega^s \omega^{-sk} |a^k \bmod N\rangle = \omega^s |u_s\rangle$$

$|u_s\rangle$ **is the eigenvector of** U_a **with eigenvalue** $\omega^s = e^{\frac{2\pi i}{r} s}$ $\quad 0 \le s < r$

Example

$N = 5$

$a = 3$

$n = \lceil \log_2 5 \rceil = 3$

$r = 4$

$\omega = e^{\frac{2\pi i}{4}}$

$$|u_s\rangle = \frac{1}{\sqrt{4}} \sum_{k=0}^{3} \omega^{-sk} |3^k \bmod 5\rangle = \frac{1}{2}\big(\underbrace{|1\rangle}_{k=0} + \underbrace{\omega^{-s}|3\rangle}_{k=1} + \underbrace{\omega^{-2s}|4\rangle}_{k=2} + \underbrace{\omega^{-3s}|2\rangle}_{k=3}\big)$$

$$U_a |u_s\rangle = \frac{1}{2}\big(|3\rangle + \omega^{-s}|4\rangle + \omega^{-2s}|2\rangle + \omega^{-3s}|1\rangle\big)$$

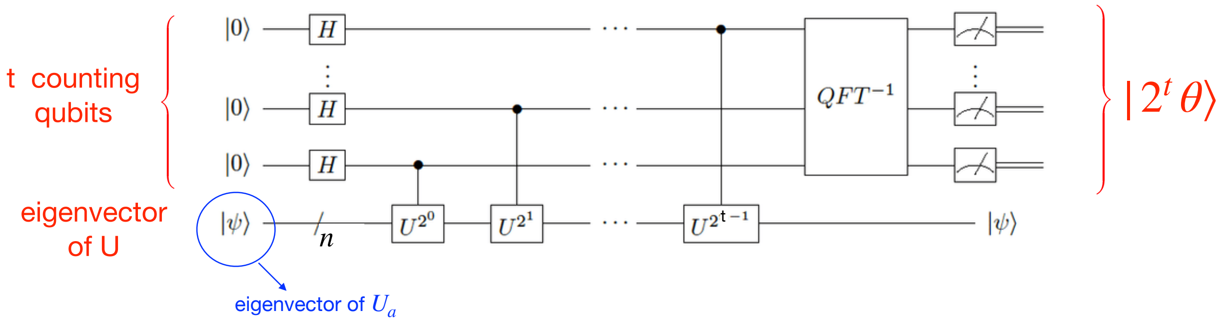
$$= \frac{1}{2}\omega^s \big(\omega^{-s}|3\rangle + \omega^{-2s}|4\rangle + \omega^{-3s}|2\rangle + \underbrace{\omega^{-4s}}_{1}|1\rangle\big)$$

$$= \omega^s |u_s\rangle = e^{\frac{2\pi i}{4} s} |u_s\rangle$$

Quantum Phase Estimation on U_a

We use QPE to estimate the eigenvalues of U_a, which encode the order r of a in the phase, with high accuracy

$$\omega^s = e^{\frac{2\pi i}{r} s} = e^{2\pi i \theta} \quad \text{for} \quad \theta = \frac{s}{r}$$



To apply QPE on U_a we need to prepare the state $|u_s\rangle$ (eigenvector of U_a)

To prepare $|u_s\rangle$ we need the order r, which we do not know…

IDEA

Instead of a single eigenvector $|u_s\rangle$, we prepare an equal superposition of all eigenvectors $|u_s\rangle$, for $0 \leq s < r$, since

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$$

and $|1\rangle = |0\rangle^{\otimes(n-1)}|1\rangle = |00\ldots01\rangle$ is a state that we can prepare easily.

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \left(\frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \omega^{-sk} |a^k \bmod N\rangle \right)$$

$$= \frac{1}{r} \sum_{s=0}^{r-1} \left(\omega^{-s\cdot 0} |a^0 \bmod N\rangle + \sum_{k=1}^{r-1} \omega^{-sk} |a^k \bmod N\rangle \right)$$

$$= \frac{1}{r} \left(\sum_{s=0}^{r-1} |1\rangle \right) + \frac{1}{r} \sum_{k=1}^{r-1} \left(\sum_{s=0}^{r-1} \omega^{-sk} \right) |a^k \bmod N\rangle$$

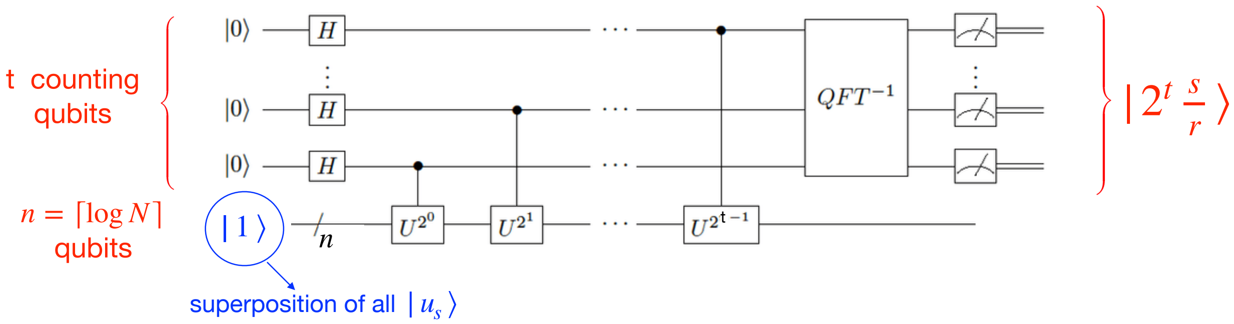
$$= |1\rangle + \frac{1}{r} \sum_{k=1}^{r-1} \left(\frac{1 - \overset{1}{\omega^{-skr}}}{1 - \omega^{-sk}} \right) |a^k \bmod N\rangle = |1\rangle$$

$\omega = e^{\frac{2\pi i}{r}}$

$|1\rangle = |0\rangle^{\otimes(n-1)}|1\rangle = |00\ldots01\rangle$

The circuit

Application of QPE on U_a using the state $|1\rangle$ to measure a phase s/r for a random integer s $(0 \leq s < r)$



TWO REGISTERS

➡ The **first register** consists of t qubits (accuracy and probability of success of QPE depend on t)

It is possible to show that the choice

$$t = 2n + 1 + \left\lceil \log\left(2 + \frac{1}{2\,\varepsilon}\right) \right\rceil \qquad n = \lceil \log N \rceil$$

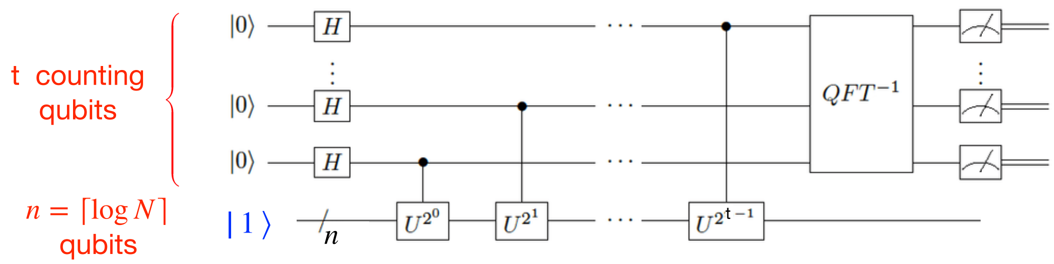
guarantees that for each s, we obtain an estimate of the phase s/r accurate to $2n + 1$ bits, with probability at least $(1 - \varepsilon)$

➡ The **second register** is prepared in state $|1\rangle$ over $n = \lceil \log N \rceil$ qubits:

$$|1\rangle \rightsquigarrow |\underbrace{00 \ldots 0}_{\text{n-1}}1\rangle = |0\rangle^{\otimes(n-1)}|1\rangle$$

n qubits are necessary to express the superposition of the eigenvectors $|u_s\rangle$ (all values are modulo N)

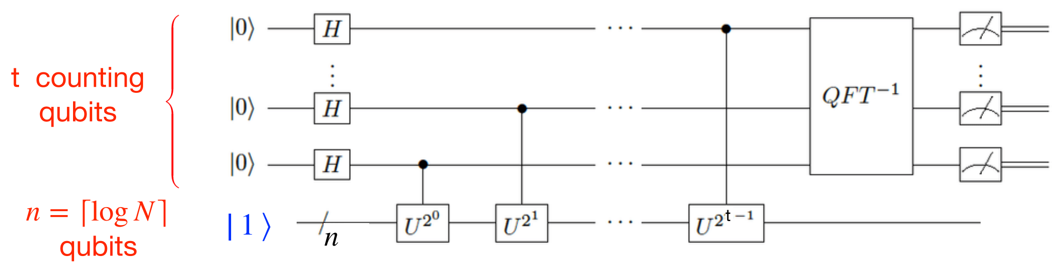
The circuit



The circuit consists of 4 steps:

1. Apply t H gates to put the first register in equal superposition
2. Apply the sequence of controlled $U_a^{2^j}$ gates, with control on the j-th qubit

The circuit



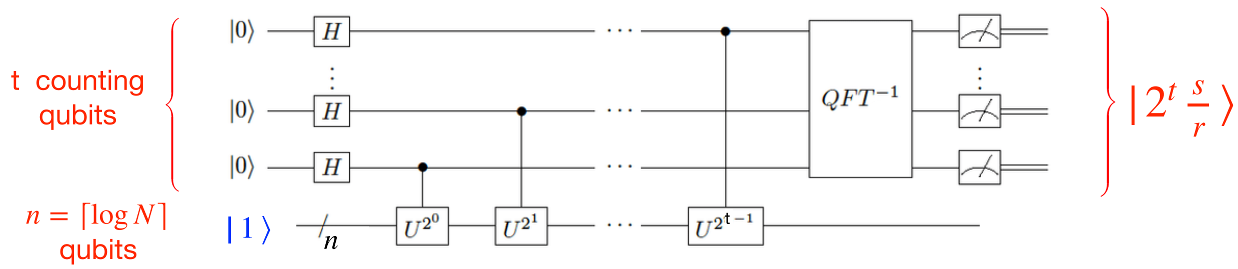
3. Apply QFT^{-1} to the first register

↳ QPE applied to a superposition of eigenvectors $|u_s\rangle$ produces a superposition of the eigenvectors $|u_s\rangle$ entangled with estimates of their eigenvalues

INPUT **OUTPUT**

$$|0\rangle^{\otimes t}|1\rangle = |0\rangle^{\otimes t}\frac{1}{\sqrt{r}}\sum_{s=0}^{r-1}|u_s\rangle \longrightarrow \frac{1}{\sqrt{r}}\sum_{s=0}^{r-1}|2^t\frac{s}{r}\rangle|u_s\rangle$$

The circuit



4. Measure the first register

↳ The uniform superposition $\displaystyle \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |2^t \frac{s}{r}\rangle |u_s\rangle$

collapses into one of the r eigenstates $|u_s\rangle$, and QPE returns an estimate of the phase s/r for a **random** (and **unknown**) integer s

↳ *The measure yields an integer x such that $\varphi = x/2^t$ is an estimate of s/r for **unknown** s and r*

We are left with two problems

1. FIND AN EFFICIENT PROCEDURE TO IMPLEMENT A CONTROLLED $U_a^{2^j}$ OPERATOR FOR ANY $0 \leq j < t$

2. OBTAIN THE ORDER r FROM THE RESULT OF THE PHASE ESTIMATION

   $\varphi \sim \dfrac{s}{r}$ (s is an unknown random integer in $[0, r-1]$)

   THIS STEP IS PERFORMED ON A CLASSICAL COMPUTER, USING AN EFFICIENT CLASSICAL ALGORITHM

   ↳ the CONTINUED FRANCTIONS ALGORITHM

1. Controlled $U_a^{2^j}$ gate implementation

The entire sequence of controlled $U_a^{2^j}$ operators can be implemented with $O(n^3)$ gates, for $n = \lceil \log N \rceil$, using MODULAR EXPONENTIATION (i.e., the square and multiply algorithm)

$$U_a^{2^j} | y \rangle = | (a^{2^j} y) \mod N \rangle = | (a^{2^j} \mod N) \cdot y \mod N \rangle$$

- ➡ We precompute $a^{2^j} \mod N,\ 0 \leq j < t,$ by repeated modular squaring of a (➔ $t-1$ squaring operations)
- ➡ Since $a^{2^j} \mod N < N$, at most n bits are needed to specify all these numbers
- ➡ The precomputed values are then used to implement the multiplications

$$((a^{2^j} \mod N) \cdot y) \mod N \qquad 0 \leq j < t$$

- ➡ Since $y < N$, a total of n bits are required to express each of the numbers

The overall complexity is $O(n^2 t) = O(n^3)$
that becomes $O(n^2 \log n \log \log n)$ using fast multiplication

2. Recover of the order from the phase

This step is performed on a classical computer.
The result of the measurement on the first register (after QPE) is

$$x = 2^t \varphi \sim 2^t \frac{s}{r}$$

We know an approximation $\varphi = x/2^t$ of s/r, with $2n+1$ bits of accuracy, i.e.

$$\left| \frac{x}{2^t} - \frac{s}{r} \right| \leq \frac{1}{2^{2n+1}} \leq \frac{1}{2N^2} \qquad \text{since } N \leq 2^n \qquad (1)$$

We also know that φ is an approximation of a rational number whose denominator r is less than N

CLAIM

There is at most one fraction s/r with $r < N$ such that

$$\left|\frac{x}{2^t} - \frac{s}{r}\right| \leq \frac{1}{2\,N^2} \qquad (1)$$

Proof
Suppose s'/r' and s''/r'' both lie within $1/(2N^2)$ of $x/2^t$, and are different. Then

$$\left|\frac{s'}{r'} - \frac{s''}{r''}\right| = \frac{|s'r'' - r's''|}{r'r''} \geq \frac{1}{r'r''} > \frac{1}{N^2} \qquad (2)$$

But s'/r' and s''/r'' are both within $1/(2N^2)$ of $x/2^t$, so they must be within $1/N^2$ of each other (triangle inequality), contradicting (2).

Hence, there is at most one s/r with $r < N$ satisfying (1).

Computation of s/r from $x/2^t$

Idea

compute the fraction nearest to $x/2^t$, with denominator less than N satisfying (1) in order to obtain r (denominator)

Bruteforce

Try out all candidate fractions s'/r' with denominator less than N, and find the unique one satisfying $\left|\frac{x}{2^t} - \frac{s'}{r'}\right| \leq \frac{1}{2\,N^2}$

↳ there are $O(N^2)$ such fractions
the method is exponential in $n = \lceil \log_2 N \rceil$

Continued Fractions Algorithm (CFA)

Classical number theory algorithm that allows to accomplish this task efficiently, in time $O(n^3)$, or $O(n^2 \log n \log \log n)$ with fast multiplication

Theory of continued fractions

The idea of continued fractions is to approximate real numbers using finite number of integers

Let $a_0, a_1, a_2, a_3, \ldots \in \mathbb{N}$

We denote with $[a_0, a_1, a_2, a_3, \ \ldots]$ the **continued fraction** defined as

$$[a_0, a_1, a_2, a_3, \ \ldots] \equiv a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \ldots}}}$$

Using a continued fraction expansion, every real number x can be expressed in terms of integer numbers, and approximated with a rational

Example

Let us approximate π to the first 4 decimal places with a rational number

$$\pi = 3{,}1415\ldots \quad = 3 + \frac{1415}{10000} \quad = 3 + \cfrac{1}{\cfrac{10000}{1415}}$$

$$10000 = 7 * 1415 + 95$$

$$= 3 + \cfrac{1}{7 + \cfrac{95}{1415}} \quad = 3 + \cfrac{1}{7 + \cfrac{1}{\cfrac{1415}{95}}} \quad = 3 + \cfrac{1}{7 + \cfrac{1}{14 + \cfrac{85}{95}}}$$

$$1415 = 14 * 95 + 85$$

$$= 3 + \cfrac{1}{7 + \cfrac{1}{14 + \cfrac{1}{\cfrac{95}{85}}}} \quad = 3 + \cfrac{1}{7 + \cfrac{1}{14 + \cfrac{1}{1 + \cancel{\frac{10}{85}}}}}$$

We consider this term small enough to be discarded

$$95 = 85 + 10$$

$$\approx 3 + \cfrac{1}{7 + \cfrac{1}{14 + 1}} = 3 + \cfrac{1}{7 + \cfrac{1}{15}} = \frac{333}{106}$$

$$CF_2(\pi) = [3, 7, 15] = \frac{333}{106} = 3.1415094339$$

Theory of continued fractions

- A continued fraction is **finite** if it is defined by a finite set of elements $a_0, a_1, \dots, a_k \in \mathbb{N}$

- A number x has a **finite** continued fractions expansion if and only if $x \in \mathbb{Q}$

$$\frac{427}{512} \equiv [0,1,5,42,2] = 0 + \cfrac{1}{1 + \cfrac{1}{5 + \cfrac{1}{42 + \frac{1}{2}}}}$$

$$\frac{1365}{8192} \equiv [0,6,682,2] = 0 + \cfrac{1}{6 + \cfrac{1}{682 + \frac{1}{2}}}$$

Example

Let us approximate the rational number 427/512

$$\frac{427}{512} = 0 + \frac{1}{\frac{512}{427}} = 0 + \cfrac{1}{1 + \frac{85}{427}} = 0 + \cfrac{1}{1 + \cfrac{1}{\frac{427}{85}}}$$

$$= 0 + \cfrac{1}{1 + \cfrac{1}{5 + \frac{2}{85}}} = 0 + \cfrac{1}{1 + \cfrac{1}{5 + \cfrac{1}{\frac{85}{2}}}}$$

$$= 0 + \cfrac{1}{1 + \cfrac{1}{5 + \cfrac{1}{42 + \frac{1}{2}}}} = [0,1,5,42,2]$$

$$512 = 427 + 85$$
$$427 = 85 * 5 + 2$$
$$85 = 42 * 2 + 1$$

The fractional part 1/2 is the reciprocal of 2, so its approximation in this scheme is exact

EXERCISE

- Verify that $\frac{415}{93} = [4,2,6,7]$
- Verify that $\varphi = \frac{1+\sqrt{5}}{2} \approx 1{,}6180339887\dots = [1,1,1,1,...]$

Theory of continued fractions

If x is an irrational number, then its continued fractions representation continues indefinitely and produces a sequence of rational approximations that converge to the starting number x as a limit

- $\sqrt{19} = [4, 2, 1, 3, 1, 2, 8, 2, 1, 3, 1, 2, 8, \ \ldots]$
  The pattern repeats indefinitely with a period of 6
  Sequence A010124 in the OEIS (On-Line Encyclopedia of Integer Sequences)
- $e = [2, 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, \ \ldots]$
  The pattern repeats indefinitely with a period of 3 except that 2 is added to one of the terms in each cycle
  Sequence A003417 in the OEIS
- $\varphi = [1, 1, 1, 1, 1, 1, 1, 1, \ \ldots]$
  The golden ratio, sequence A000012 in the OEIS
- $\pi = [3, 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, \ \ldots]$
  No pattern has ever been found in this representation
  Sequence A001203 in the OEIS

Theory of continued fractions

Definition

Let $CF(x) = [a_0, a_1, a_2, \ \ldots], \ a_i \in \mathbb{N}^+$, be the continued fraction representation of $x \in \mathbb{R}$

The k**-th convergent** of x is the rational number

$$CF_k(x) = [a_0, a_1, \ldots a_k] = p_k \big/ q_k$$

Lemma

$CF_k(x) = [a_0, a_1, \ldots a_k] = p_k \big/ q_k$ is the **best rational approximation** of x with denominator at most q_k

↳ p_k / q_k is closer to x than any approximation with a smaller or equal denominator

Example

$$CF_k(x) = [a_0, a_1, \ldots a_k] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ldots + \frac{1}{a_k}}}} = \frac{p_k}{q_k}$$

$$\frac{427}{512} \equiv [0, 1, 5, 42, 2] = 0 + \cfrac{1}{1 + \cfrac{1}{5 + \cfrac{1}{42 + \frac{1}{2}}}}$$

The convergents are

- $\frac{p_0}{q_0} = 0$
- $\frac{p_1}{q_1} = 0 + \frac{1}{1} = 1$
- $\frac{p_2}{q_2} = 0 + \cfrac{1}{1 + \frac{1}{5}} = \frac{5}{6}$
- $\frac{p_3}{q_3} = 0 + \cfrac{1}{1 + \cfrac{1}{5 + \frac{1}{42}}} = \frac{211}{253}$
- $\frac{p_4}{q_4} = 0 + \cfrac{1}{1 + \cfrac{1}{5 + \cfrac{1}{42 + \frac{1}{2}}}} = \frac{427}{512}$

Example

$$CF_k(x) = [a_0, a_1, \ldots a_k] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ldots + \frac{1}{a_k}}}} = \frac{p_k}{q_k}$$

$$\pi \approx 3{,}1415926535\ldots \equiv [3, 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, \ \ldots]$$

The convergents are

$$3, \frac{22}{7}, \frac{333}{106}, \frac{355}{113}, \frac{103993}{33102}, \ldots$$

- $\frac{p_0}{q_0} = 3$
- $\frac{p_1}{q_1} = 3 + \frac{1}{7} = \frac{22}{7} = 3.1428571428$
- $\frac{p_2}{q_2} = \frac{333}{106} = 3.1415094339$
- $\frac{p_3}{q_3} = \frac{355}{113} = 3.1415929203$
- $\frac{p_4}{q_4} = \frac{103993}{3102} = 3.1415926530$

Remarks

Given a rational number φ, the sequence of the convergents of its continued fractions expansion $CF(\varphi)$ has the following properties:

- the convergents are in their lowest terms
- with increasing k, the difference between φ and its k-th convergent gets smaller and smaller (i.e., the convergents form an approximation of φ that gets better and better)
- the denominator of the convergents are increasing

The convergent can be used to approximate the rational number φ by fractions with smaller denominator

Algorithm CFA

There exists an algorithm that efficiently computes the k-th convergent

Let $p/q \in \mathbb{Q}$

If p and q (reduced to lowest terms) are n-bit integers, then

- the length of the continued fraction expansion is $O(n)$
- the continued fraction expansion and its convergents can be computed in time $O(n^3)$, or $O(n^2 \log n \log \log n)$ with fast multiplication

The best rational approximation

THEOREM 2

Let $\varphi \in \mathbb{Q}$ and suppose that $\dfrac{s}{r} \in \mathbb{Q}$ is such that

$$\left|\varphi - \frac{s}{r}\right| \leq \frac{1}{2\, r^2}$$

Then $\dfrac{s}{r}$ is a convergent of the continued fraction expansion of φ

Example

Let $\varphi = \dfrac{427}{512}$ be an approximation of an unknown rational number $\dfrac{s}{r}$ such that $r < 21$ and

$$\left|\varphi - \frac{s}{r}\right| \leq \frac{1}{2 \cdot 21^2}$$

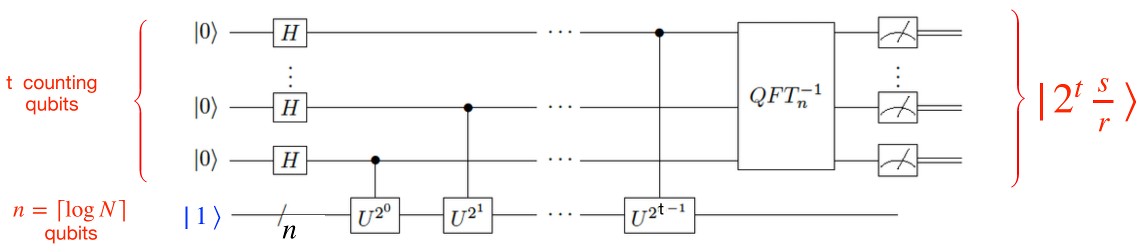
Consider $CF(\varphi) = [0, 1, 5, 42, 2]$ and the convergents

$$0, 1, \frac{5}{6}, \frac{211}{253}, \frac{427}{512}$$

We get

$$\frac{s}{r} = \frac{5}{6}$$

2. Recover of the order from the phase



Let us return to the problem of getting the order r from the knowledge of $\varphi = x/2^t$, where x is the value measured on the first register, satisfying

$$\left|\varphi - \frac{s}{r}\right| \leq \frac{1}{2\,N^2} \quad \text{and} \quad r < N$$

We know that there is (at most) a unique fraction s/r such that

$$\left|\varphi - \frac{s}{r}\right| \leq \frac{1}{2\,N^2} \qquad \text{(see Claim 1)}$$

Since $r < N$, we have $\left|\varphi - \frac{s}{r}\right| \leq \frac{1}{2\,N^2} < \frac{1}{2\,r^2}$

Thus, Theorem 2 applies: the fraction s/r must be a **convergent** of the continued fraction expansion of $\varphi = x/2^t$

We run the Continued Fraction Algorithm

- We compute the convergents of $\varphi = x/2^t$, and check through the list of convergents to find the **unique** one satisfying the requirements

$$\left|\varphi - \frac{s}{r}\right| \leq \frac{1}{2\,N^2} \quad \text{and} \quad r < N$$

- Since x and 2^t are $O(n)$ bit integers, the number of convergents is $O(n)$, and they can be computed in time $O(n^3)$, or $O(n^2 \log n \log \log n)$ with fast multiplication
- Thus, CFA produces two numbers s' and r' without common factors, s.t.

$$\frac{s'}{r'} = \frac{s}{r}$$

- $2n + 1$ bits of accuracy guarantee that we can determine s'/r' uniquely from $\varphi = x/2^t$
- r' is the candidate for the order, thus we check whether $a^{r'} \bmod N = 1$

REMARK: in order to get r, s and r must be coprime

Complexity

If s and r are n-bit integers, we can compute s'/r' using $O(n)$ steps, each of cost $O(n^2)$, or $O(n \log n \log \log n)$ with fast multiplication